

Why Distributed Systems?

Distributed Systems

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Pervasive Computation & Interaction I

Nowadays, computational systems. . .

- . . . have become **pervasive**, since they are everywhere, and tend to affect every aspect of our everyday life and activity
- . . . are at the core of most (if not all) *artificial systems*, so that every principled discipline for modelling / engineering computational systems affects the *modelling and engineering of almost every sort of artificial systems*

Pervasive Computation & Interaction II

Pervasiveness of computations

- we live immersed in a sort of ever-expanding *computational bubble*, where an huge number of computations are performed at every instant around us
 - at home, in our cars, in the workplaces, in hospitals, in airports and train stations, in schools and universities, in public spaces, . . .
- **distributed**, *concurrent* computations
 - either controlled / triggered, or **autonomous** computations

Pervasiveness of interaction

- almost any computational system of today comes equipped with ICT technologies for **interacting** with other computational systems
- computational devices *continuously interact*
 - with humans
 - with each others
 - with the physical *environment* and its *resources*



Physical vs. Computational

The physical nature of artificial systems...

... adds complexity to computational components / systems

- in terms of **distribution**
 - **spatial** *distribution*
 - **temporal** *distribution*
- in terms of **unpredictability** of the *environment* where they have to work

On the Notion of Distribution I

What is *spatially* distributed?

- ① *computational* units
- ② *communication* channels
- ③ *data / information / knowledge*
 - along with their *representations*
- ④ sensors, actuators, ...
 - the boundary between the system and the surrounding *environment* is *spatially sparse*

On the Notion of Distribution II

What is *temporally* distributed?

- basically, **events**
 - things happens, yet no longer in a clearly-ordered sequence
- events are scattered, and the trivial *before/after* temporal relation simply cannot be applied to many pairs of events

Spatio-temporal unity of systems is lost

- there is no longer a notion of *system time*, nor a system *location*
- system components, at different level of abstraction, are only *partially correlated*
 - temporally & spatially

On the Notion of Distribution III

A number of assumptions over systems no longer hold

- system **events** *no longer* constitute a totally-ordered set
 - generally speaking, *partial ordering* is the only feature
- admissible **interactions** among system components *no longer* depend on *compresence*
 - in space / time
 - within the same physical / virtual *topology*



So, Why Distributed Systems? I

Centralised vs. Distributed Systems^[Puder et al., 2005]

Criteria	Centralized system	Distributed system
Economics	low	high
Availability	low	high
Complexity	low	high
Consistency	simple	difficult
Scalability	poor	good
Technology	homogenous	heterogenous
Security	high	low

So, Why Distributed Systems? II

We need distributed systems for^[Ghosh, 2014]

- geographically distributed environments
- computation speedup
- resource sharing
- fault tolerance



Artificial Systems Nowadays I

Conceiving and constructing artificial systems...

- ... nowadays means dealing with *distributed systems*
- whose core is represented by (distributed) *computational systems*
- which are to be *modelled* and *built*



Artificial Systems Nowadays II

Modelling distributed (artificial) systems...

- ... involves new (theoretical) problems
- so, it requires new theoretical frameworks, models, abstractions, techniques
- mostly, computational ones
- ! it is one of the main objects of study of **computer science**



Artificial Systems Nowadays III

Building distributed (artificial) systems. . .

- . . . involves new (practical) problems
- so, it requires new technologies, infrastructures, methods, methodologies
- mostly, computational ones
- ! it is one of the main objects of study of **computer engineering**

Artificial Systems Nowadays IV

As a result...

- ... we will mix up computer science and engineering issues
 - we may also say, theoretical and methodological / technological issues all along the course
- starting from the very beginning of the course

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